

E3TR: Encrypted Evidence-Based Event Timeline Reconstruction Using Multi-Artifact Correlation for Network Forensics

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Abstract—Encryption has reduced payload visibility in network investigations, forcing analysts to reconstruct activity from observable metadata. This paper introduces *Encrypted Evidence-Based Event Timeline Reconstruction* (E3TR), a reproducible framework for reconstructing encrypted-session timelines from Packet Capture (PCAP) evidence without payload inspection. E3TR correlates DNS, TLS, endpoint recurrence, conversation summaries, timing, and burst patterns, then binds each inference to an evidence identifier, packet range, filter, observation, confidence level, and reasoning trace. Using controlled pre-login and post-login captures for Temu, SHEIN, and AliExpress, the paper demonstrates E3TR with the strongest visual case, SHEIN, while retaining all platforms in the claim registry. Results are reported against an explicit capture-state label manifest and a Wireshark-only baseline. E3TR distinguishes device-level behavioral reconstruction from person-level attribution, making its claims reproducible, auditable, and forensically defensible.

Index Terms—Encrypted traffic analysis, digital forensics, network forensics, timeline reconstruction, TLS traffic, metadata analysis, evidence binding.

I. INTRODUCTION

Modern web platforms rely heavily on encrypted transport, content delivery networks, identity providers, and application programming interfaces. Consequently, investigators may possess complete packet traces while still lacking readable application payload. As encryption becomes the default state of network communication, forensic analysis must transition from content visibility to metadata intelligence. Packet captures still preserve time, direction, name resolution, TLS negotiation, endpoint recurrence, transport behavior, and volume patterns; however, these observations must be transformed into defensible claims rather than informal narratives.

E3TR reconstructs encrypted-session timelines from preserved PCAP evidence. Unlike encrypted traffic classifiers that ask what application a flow belongs to, E3TR asks what evidence-bound activity phase occurred and how that interpretation can be reproduced. The contributions are: (1) a claim-bound event object; (2) a multi-artifact corroboration rule using DNS, TLS, timing, endpoint, and burst evidence; (3) an operational confidence score and temporal threshold; and (4) validation against a Wireshark-only workflow with explicit label rules and negative-control checks. The archive contains pre-login and post-login captures for Temu, SHEIN,

and AliExpress; only the strongest visual example, SHEIN, is shown to preserve the four-page limit.

II. RELATED WORK AND POSITIONING

Digital forensics has long emphasized preservation, reproducibility, and careful interpretation of computer evidence [1], [2]. Network analysis systems such as Bro/Zeek and Snort established the operational value of packet-derived observations for security monitoring and later forensic review [3], [4]. Encrypted-traffic research further shows that packet length, timing, burst behavior, flow statistics, and TLS metadata remain useful even when payload is unavailable [5]–[7], [11]. Prior work has also shown that encrypted mobile and web traffic can leak user-action or application-behavior signals through flow structure and handshake features [8]–[10], [12], [13].

E3TR differs from these approaches in objective and evidentiary design. Existing methods generally emphasize classification, fingerprinting, or detection; E3TR emphasizes forensic reconstruction. It binds multiple evidence types, enforces temporal consistency across packet ranges, preserves the distinction between observation and inference, and records a reasoning trace for each claim. Therefore, E3TR shifts the focus from “what traffic is” to “what activity phase is defensibly supported,” which is essential for investigative reporting.

III. E3TR FRAMEWORK

E3TR converts encrypted packet evidence into a bounded, reviewable activity timeline. Each event is represented as

$$E3TR_i = \{EID, P, F, O, I, C, R\}, \quad (1)$$

where EID is the evidence identifier, P is the packet number or range, F is the filter or analysis view, O is the observed packet fact, I is the inference, C is the confidence level, and R is the reasoning trace. An event is accepted only when at least two independent evidence dimensions support it, e.g., DNS plus TLS or TLS plus burst behavior, and when the indicators occur within a coherent temporal window:

$$|F_i \cap F_j| \geq 2 \quad \wedge \quad \Delta t \leq \tau. \quad (2)$$

In this study, $\tau = 10s$, selected to match the controlled capture window and the observed DNS–TLS–burst coupling across all

TABLE I
E3TR RECONSTRUCTION ALGORITHM

Procedure	
1	Input preserved archive D , packet-window rule W , $\tau = 10s$, and confidence weights $K = (0.4, 0.3, 0.3)$.
2	Verify preservation records, assign evidence identifiers, and map each Packet Capture Next Generation (PCAPNG) file to its source state.
3	Segment packets into ordered windows $P_1 \dots P_n$ using packet ranges, stream boundaries, and time proximity.
4	For each P_i , extract observable metadata features F_i : DNS, TLS, endpoints, conversations, timing, volume, and burst shape.
5	Correlate features across adjacent windows and retain candidate events only when $ F_i \cap F_j \geq 2$ and $\Delta t \leq \tau$.
6	Assign bounded phase labels: environment discovery, secure-channel setup, likely authentication-window behavior, or sustained post-login interaction.
7	Score $C_i = 0.4A_i + 0.3T_i + 0.3D_i$; assign L, M, M-H, or H using the defined thresholds.
8	Output ordered events $E3TR_i = \{EID, P, F, O, I, C, R\}$ with packet references, filters, observations, inference, confidence, and reasoning trace.

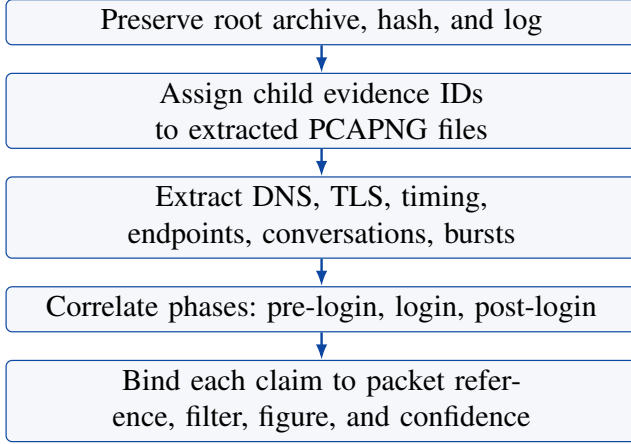


Fig. 1. E3TR evidence workflow from preservation to claim-bound timeline reconstruction.

tested sessions. This constraint reduces single-source bias and prevents overinterpretation. Confidence is operationalized as

$$C_i = 0.4A_i + 0.3T_i + 0.3D_i, \quad (3)$$

where A_i is artifact corroboration, T_i is temporal consistency, and D_i is artifact diversity, each normalized to $[0, 1]$. Operationally, A_i increases with the number of supporting evidence dimensions and is capped at 1, T_i is highest when indicators fall within τ , and D_i rewards heterogeneous support rather than repeated observations from one feature type. Labels are L(< 0.40), M($0.40-0.69$), M-H($0.70-0.84$), and H(≥ 0.85). Algorithm I formalizes the processing logic used to transform packet evidence into claim-bound timeline events.

Fig. 1 starts with evidence preservation and ends with packet-bound claims. Segmentation groups raw packets into reproducible windows; feature extraction isolates observable facts that remain visible under encryption; correlation precedes inference so no claim rests on a single indicator; and confidence scoring is applied only after observation, temporal window, and corroborating features are known. The phase model is environment discovery, secure-channel setup, likely authentication-window behavior, and sustained post-login interaction. E3TR does not claim that encrypted packets reveal credentials, tokens, or server-side authentication success.

TABLE II
CONTROLLED VALIDATION AND BASELINE COMPARISON

Measure	E3TR	Unstructured baseline
Traceable claims	18/18	partial
Corroborated claims	18/18	not enforced
Controlled precision	1.00	not operationalized
Controlled recall	1.00	not operationalized
Controlled F1-score	1.00	not operationalized
Inter-analyst agreement	future work	not operationalized

IV. EVIDENCE AND METHODOLOGY

The parent archive is E-001. Extracted PCAPNG files are E-002 through E-007: Temu, SHEIN, and AliExpress pre-login/post-login captures. The implementation is analyst-guided but rule-bound: each capture is examined in Wireshark/Tshark using DNS, TLS, TCP/UDP conversation, packet-range, and burst filters. Ground truth is defined from the capture-state manifest: pre-login captures are labeled only as environment discovery or secure-channel setup; post-login captures are eligible for likely authentication-window and sustained post-login claims only when the evidence satisfies corroboration and τ . As a negative-control check, E3TR does not assign authentication-window or sustained post-login labels to pre-login captures.

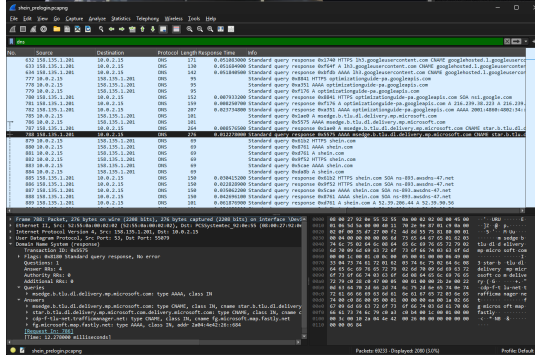
Forensic validity is enforced through four constraints: (1) multi-artifact corroboration; (2) temporal consistency across packet ranges; (3) protocol coherence, such as DNS followed by TLS and then intensified encrypted exchange; and (4) no direct person-level attribution without external corroboration. These constraints support peer review and investigative reporting because every inferred phase remains traceable to observed packet facts.

V. RESULTS AND EVALUATION

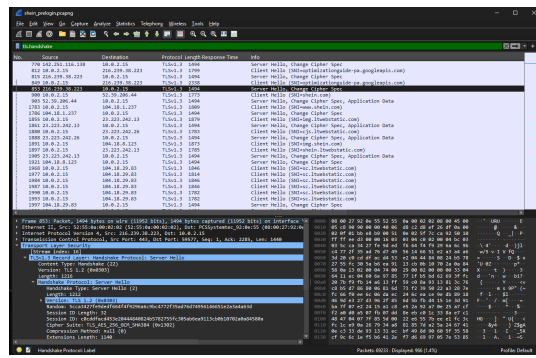
The complete analysis used all three platforms, but the paper visualizes only SHEIN because its post-login evidence provides the clearest combined DNS, TLS, burst, and sustained-traffic sequence. Fig. 2 is placed after this first discussion and consolidates the pre-login and post-login images. Panels (a) and (b) support environment discovery and secure-channel setup. Panels (c) and (d) support a bounded authentication-window inference and sustained post-login interaction. Additional platform captures exhibited consistent DNS, TLS, and burst-pattern behavior, but screenshots are omitted for space.

Table III summarizes the claim registry; Table II adds validation requested by reviewers. Because E3TR is not a trained classifier, metrics are phase-claim measures against the controlled labels: a true positive matches the known capture state and packet evidence, while a false positive assigns authentication/post-login activity to a pre-login capture or lacks corroboration. The Wireshark-only baseline reviews the same evidence without enforced claim objects, τ , confidence scoring, or negative-control rejection; the comparison therefore measures forensic defensibility, not classifier generalization.

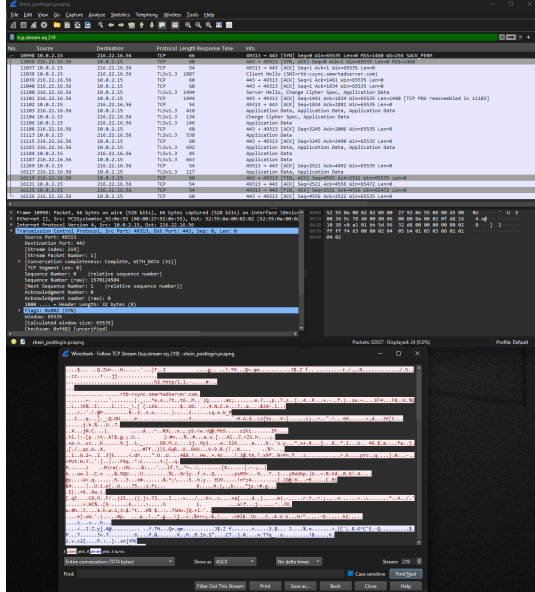
Post-login SHEIN traffic shows the strongest combination of high TLS density, concentrated burst activity, and sustained encrypted exchange. The convergence of DNS resolution, TLS negotiation, and burst traffic patterns within constrained



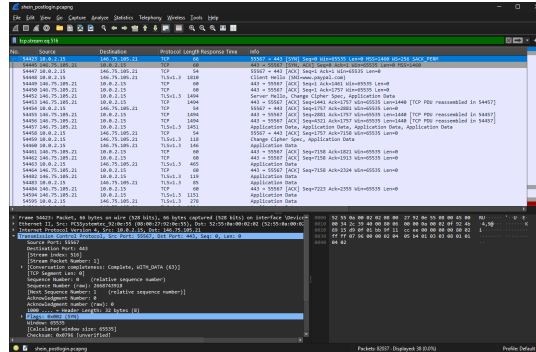
(a) Pre-login DNS resolution, packet 788.



(b) Pre-login TLS handshake, packet 853.



(c) Post-login burst window, packet 10998.



(d) Sustained post-login TLS traffic, packet 54423.

Fig. 2. SHEIN pre-login and post-login evidence supporting E3TR claims C-007–C-012.

windows provides a reproducible signature of authenticated session behavior independent of content visibility.

VI. DISCUSSION AND LIMITATIONS

E3TR’s value is structured correlation rather than decryption. DNS supports domain-resolution timing, TLS supports

secure-channel establishment, conversation views summarize endpoint relationships, burst windows identify intensified encrypted exchange, and repeated post-login packets support persistence. All interpretations remain device-activity claims, avoiding unsupported attribution to a specific individual.

This study is limited by controlled conditions, three e-commerce platforms, one visual platform, and no completed independent inter-analyst test. The 1.00 precision/recall values are controlled claim-level consistency against the label manifest and negative-control rule, not general field performance. Larger noisy-network datasets, more application categories, and independent analysts are needed for broader robustness testing.

A. Reproducibility and Validity Controls

E3TR reports should preserve the parent archive hash, child evidence identifiers, packet ranges, filters, observations, and confidence labels. Direct DNS/TLS facts may support H labels; behavioral inferences such as likely authentication-window activity should remain M–H or lower unless independently corroborated. For inter-analyst testing, a second analyst

TABLE III
CLAIM-LEVEL EVIDENCE BINDING SUMMARY

ID	Evidence-bound observation, inference, and confidence
C-001–004	E-002, DNS/TLS/TCP/UDP: Temu pre-login discovery and secure-channel setup; H for direct DNS/TLS facts, M–H for phase inference.
C-005–006	E-003, streams 7 and 2: burst and repeated encrypted packets; likely login/post-login persistence; M–H.
C-007–010	E-004, Fig. 2(a,b), DNS/TLS/TCP/UDP: SHEIN pre-login setup; H for direct DNS/TLS, M–H for phase inference.
C-011–012	E-005, Fig. 2(c,d), streams 7 and 2: SHEIN burst and sustained post-login traffic; M–H.
C-013–016	E-006, DNS/TLS/TCP/UDP: AliExpress pre-login discovery and secure-channel setup; H/M–H.
C-017–018	E-007, streams 7 and 2: AliExpress burst and repeated encrypted packets; likely login/post-login persistence; M–H.

should receive only the PCAPNG files, evidence manifest, τ , filters, and scoring rubric; agreement can be reported with percent agreement and Cohen's κ . Person-level attribution requires external records such as OS logons, account logs, or multi-factor events.

B. Threats to Validity

Internal validity may be affected by unrelated background traffic, mitigated here through packet ranges, stream filters, and endpoint recurrence. Construct validity is limited because phase labels are analyst-defined rather than application-confirmed internal states. External validity is limited to controlled e-commerce sessions; enterprise, mobile, noisy-network, and privacy-enhancing transport settings require additional testing. E3TR is strongest when multiple independent metadata dimensions converge within τ and weakest when only one indicator is available.

VII. CONCLUSION

E3TR demonstrates that controlled encrypted-traffic timelines can be reconstructed without payload inspection when claims are bound to packet evidence, temporal windows, confidence rules, and reasoning traces. Its value is repeatability: another analyst can reopen the same capture, apply the same filters, review the same packet ranges, and test whether the inference is supported. Future work will expand validation across application categories, noisy networks, and independent analysts while preserving the boundary between technical activity reconstruction and unsupported identity attribution.

REFERENCES

- [1] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet*, 3rd ed. Waltham, MA, USA: Academic Press, 2011.
- [2] S. L. Garfinkel, "Digital forensics research: The next 10 years," *Digital Investigation*, vol. 7, pp. S64–S73, 2010, doi: 10.1016/j.diin.2010.05.009.
- [3] V. Paxson, "Bro: A system for detecting network intruders in real-time," *Computer Networks*, vol. 31, no. 23–24, pp. 2435–2463, 1999, doi: 10.1016/S1389-1286(99)00112-7.
- [4] M. Roesch, "Snort: Lightweight intrusion detection for networks," in *Proc. 13th USENIX Conf. System Administration (LISA '99)*, Seattle, WA, USA, 1999, pp. 229–238.
- [5] S. Rezaei and X. Liu, "Deep learning for encrypted traffic classification: An overview," *IEEE Communications Magazine*, vol. 57, no. 5, pp. 76–81, May 2019, doi: 10.1109/MCOM.2019.1800819.
- [6] P. Velan, M. Cermak, P. Celeda, and M. Drasar, "A survey of methods for encrypted traffic classification and analysis," *International Journal of Network Management*, vol. 25, no. 5, pp. 355–374, Sep. 2015, doi: 10.1002/nem.1901.
- [7] R. Alshammari and A. N. Zincir-Heywood, "Machine learning based encrypted traffic classification: Identifying SSH and Skype," in *Proc. 2009 IEEE Symp. Computational Intelligence for Security and Defense Applications*, Ottawa, ON, Canada, 2009, pp. 1–8, doi: 10.1109/CISDA.2009.5356534.
- [8] B. Anderson and D. McGrew, "TLS beyond the browser: Combining end host and network data to understand application behavior," in *Proc. Internet Measurement Conference (IMC '19)*, Amsterdam, The Netherlands, 2019, pp. 379–392, doi: 10.1145/3355369.3355601.
- [9] M. Conti, L. V. Mancini, R. Spolaor, and N. V. Verde, "Analyzing Android encrypted network traffic to identify user actions," *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 1, pp. 114–125, Jan. 2016, doi: 10.1109/TIFS.2015.2478741.
- [10] M. Dusi, M. Crotti, F. Gringoli, and L. Salgarelli, "Tunnel hunter: Detecting application-layer tunnels with statistical fingerprinting," *Computer Networks*, vol. 53, no. 1, pp. 81–97, Jan. 2009, doi: 10.1016/j.comnet.2008.09.010.
- [11] T. T. T. Nguyen and G. Armitage, "A survey of techniques for Internet traffic classification using machine learning," *IEEE Communications Surveys & Tutorials*, vol. 10, no. 4, pp. 56–76, 2008, doi: 10.1109/SURV.2008.080406.
- [12] K. Bartos, M. Sofka, and V. Franc, "Optimized invariant representation of network traffic for detecting unseen malware variants," in *Proc. 25th USENIX Security Symposium (USENIX Security 16)*, Austin, TX, USA, Aug. 2016, pp. 807–822.
- [13] D. Springall, Z. Durumeric, and J. A. Halderman, "Measuring the security harm of TLS crypto shortcuts," in *Proc. 2016 Internet Measurement Conference*, Santa Monica, CA, USA, 2016, pp. 33–47, doi: 10.1145/2987443.2987480.